

of the finger and stores the template in its memory.

During matching of the fingerprint, the system generates a template when the user places his finger on the optical sensor and compares it with the templates already stored in the library. If a match is found, the user is verified.

There are certain protocols to be followed for successful communication between the module and the PC. Fingerprint detection, pattern storage and image pattern retrieval involve complex steps. The secret lies in the proper usage of different instruction codes defined for this module. You can refer the instruction code table given in the datasheet of R305 module.

Communication between the module and the PC is done in semiduplex asynchronous, serial mode. The default baud rate is 57,600 bps. You may set the baud rate to 9600-115,200 bps.

Data is transferred with frame size of 10 bits, which include the low-level starting bit, 8-bit data with the least significant bit (LSB) first and an ending bit. There is no check bit. At power-on, it takes about 500 ms for initialisation. During this period, the module cannot accept commands from your PC.

When communicating with the PC, commands sent and data received are all wrapped in package format. Commands are sent from the PC to the module, which then acknowledges the commands. Communication process requires a particular sequence to be followed as explained below.

First, the user should enroll his fingerprint by entering a password. The system will collect the fingerprint using the following code:

```
writebuff = new byte[] { 0xEF, 0x01,
0xFF, 0xFF, 0xFF, 0xFF, 0x01, 0x00,
0x03, 0x01, 0x00, 0x05 };
port.Write(writebuff, 0, writebuff.Length);
```

This information has to be entered into character buffer using the following code:

```
writebuff = new byte[] { 0xEF, 0x01,
0xFF, 0xFF, 0xFF, 0xFF, 0x01, 0x00,
0x04, 0x02, 0x01, 0x00, 0x08 };
port.Write(writebuff, 0, writebuff.Length);
```

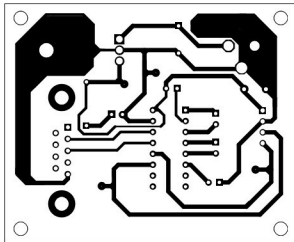


Fig. 6: An actual-size, single-side PCB layout for PC-based fingerprint scanner

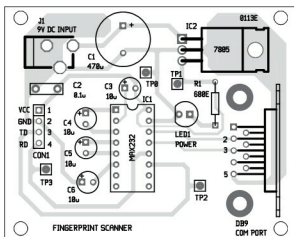


Fig. 7: Component layout for the PCB

Length);

The next step is to generate a template using the following code:

```
writebuff = new byte[] { 0xEF, 0x01,
0xFF, 0xFF, 0xFF, 0xFF, 0x01, 0x00,
0x03, 0x05, 0x00, 0x09 };
port.Write(writebuff, 0, writebuff.Length);
```

Now the template has to be stored in the memory. Once it is stored, next time when the user places his finger on the module, the system will recognise his fingerprint pattern.

Construction and testing

An actual-size, single-side PCB layout for the PC-based fingerprint scanner is shown in Fig. 6 and its component layout in Fig. 7.

Connect the circuit as shown in Fig. 3. Connect the circuit to COM port of your PC using standard RS-232 cable. Make sure that your COM port is working well. Now run the fingerprint.

TABLE III
Test Points

Test point	Details
TP0	0V, ground
TP1	5V
TP2	Sequence of pulses
TP3	Sequence of pulses

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on www.efymag.com website.

exe program. Select COM port from the drop-down menu.

Before using the system, enroll your fingerprint by pressing 'Enroll' button followed by entering a password. The default password is 'efy123'. To enroll, place your finger on the optical sensor twice. The system will process the fingerprint image to generate a template and store the template in its memory.

If enrollment completes successfully, message 'Enrolled Successfully!' will appear on the screen. Next time, when you place your finger on the sensor module, it will display 'Allowed.' If you place your finger on the sensor without enrolling, the screen will display message 'Not Allowed.'

Troubleshooting the circuit is simple. First, check the power supply between TP0 and TP1 test points. It should be 5V. The voltage status at various test points in the circuit is listed in Table III. TP2 and TP3 test points cannot be checked using a multimeter. You need an oscilloscope to check the signal pulses at these test points. During communication between the PC and R305 module, you can observe on the oscilloscope a sequence of pulses from these test points. ●

Sani Theo is a B.Tech in electronics & communication and Lalit Prakash Vatsal is a third-year integrated dual-degree student at IIT Roorkee